

EXPERIMENT 08:

8. Construct a C program to simulate Round Robin scheduling algorithm with C.

PSEUDO CODE :

Start

Input number of processes

Input Burst Time

Input Time Quantum

While processes remain

Execute each process for Time Quantum

If process is unfinished

Put it back in Ready Queue

Else

Mark completed

Calculate Waiting Time

Calculate Turnaround Time

Display results

Stop

Time Quantum = 2

Process BT WT TAT

P1 5 7 12

P2 3 6 9

P3 4 7 11

Average Waiting Time = 6.67

Average Turnaround Time = 10.67

CODE:

```
#include <stdio.h>

int main()
{
    int n,i,remain,time=0,quantum;
    int bt[20],rt[20];

    printf("Enter Number of Processes: ");
    scanf("%d",&n);

    remain=n;

    for(i=0;i<n;i++)
    {
        printf("Burst Time of P%d: ",i+1);
        scanf("%d",&bt[i]);
        rt[i]=bt[i];
    }

    printf("Enter Time Quantum: ");
    scanf("%d",&quantum);

    while(remain!=0)
    {
        for(i=0;i<n;i++)
        {
            if(rt[i]>0)
            {
                if(rt[i]<=quantum)
                {
                    time+=rt[i];
                    rt[i]=0;
                }
            }
        }
    }
}
```

```

        remain--;

        printf("P%d Completed at %d\n",i+1,time);
    }

    else

    {

        rt[i]-=quantum;

        time+=quantum;

    }

}

}

}

return 0;

}

```

The screenshot displays the CodeChef Online C Compiler interface. The top navigation bar includes links for Courses, Practice, Compete, and Compiler, along with options to Upgrade to Pro, Login, and Sign Up. The main area is titled "Online C Compiler" and features a "Practice C problems from our roadmap" link.

The code editor shows a C program with the following content:

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int n,i,remain,time=0,quantum;
6     int bt[20],rt[20];
7
8     printf("Enter Number of Processes: ");
9     scanf("%d",&n);
10
11     remain=n;
12
13     for(i=0;i<n;i++)
14     {
15         printf("Burst Time of P%d: ",i+1);
16         scanf("%d",&bt[i]);
17         rt[i]=bt[i];
18     }
19
20     printf("Enter Time Quantum: ");
21     scanf("%d",&quantum);
22
23     while(remain!=0)
24     {
25         for(i=0;i<n;i++)
26         {
27             if(rt[i]>0)
28             {
29                 if(rt[i]<=quantum)

```

The output section shows the following results:

```

>
P1 3
P2 4
P3 4

```

The status bar indicates "Status : Time limit exceeded" with an "AI Explain" button. The execution details show "Time: 5.0000 secs" and "Memory: 1.712 Mb". The sample input section displays:

```

3
P1 3
P2 4
P3 4

```